

Advanced NLP

Session 06 · The Story of LLMs

Prof. Dr. Richard Sieg · TH Köln IWS · WS 26 / 27
Wednesday, October 14, 2026

From GPT-1 to reasoning models in one hour

Time

11:15	GPT: from translation to a pretrained decoder
11:30	Scaling: bigger, more data, and how much of each
11:40	Assistants: InstructGPT and ChatGPT
11:45	Today: open weights, the architecture drift, reasoning
12:15	Lunch

Now

Tue 13

01 Kick-off

02 Attention I

Lunch

03 Transformer

04 Generation

Wed 14

05 Pretraining

06 Story of LLMs

Lunch

07 Post-training

08 Evaluation

Thu 15

Hackathon

Lunch

Presentations

9:30

17:00

Yesterday's timeline ends where this story begins

~1950s–1990s

~early 2000s

2013

~2014–2017

2017

2018–present

Statistical LMs
n-gram and Markov
models

Feedforward neural
LMs
networks that learn
word embeddings
jointly

word2vec
efficient dense word
embeddings

RNNs and LSTMs
better sequence
handling

Transformer
attention changes
language modeling

**Large language
models**
BERT, GPT, T5 and
more

Every LLM since GPT-1 is your model, with a few lines changed.

Agenda

01. GPT, 2017 to 2020

02. Scaling, 2020 to 2022

03. Assistants, 2022

04. Today, 2023 to 2026

01.

GPT, 2017 to 2020

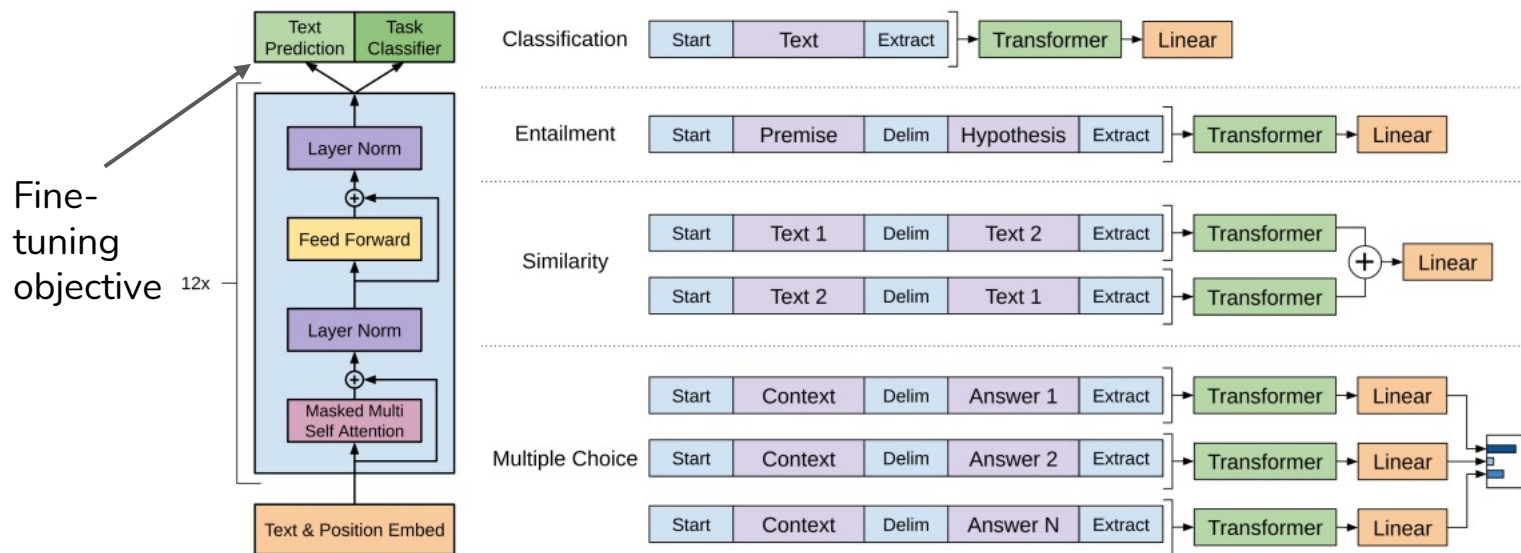
The Transformer won translation first

Model	BLEU		Training Cost (FLOPs)	
	EN-DE	EN-FR	EN-DE	EN-FR
ByteNet [18]	23.75			
Deep-Att + PosUnk [39]		39.2		$1.0 \cdot 10^{20}$
GNMT + RL [38]	24.6	39.92	$2.3 \cdot 10^{19}$	$1.4 \cdot 10^{20}$
ConvS2S [9]	25.16	40.46	$9.6 \cdot 10^{18}$	$1.5 \cdot 10^{20}$
MoE [32]	26.03	40.56	$2.0 \cdot 10^{19}$	$1.2 \cdot 10^{20}$
Deep-Att + PosUnk Ensemble [39]		40.4		$8.0 \cdot 10^{20}$
GNMT + RL Ensemble [38]	26.30	41.16	$1.8 \cdot 10^{20}$	$1.1 \cdot 10^{21}$
ConvS2S Ensemble [9]	26.36	41.29	$7.7 \cdot 10^{19}$	$1.2 \cdot 10^{21}$
Transformer (base model)	27.3	38.1	$3.3 \cdot 10^{18}$	
Transformer (big)	28.4	41.8	$2.3 \cdot 10^{19}$	

	Base	Big
Layers in Encoder & Decoder	6	6
#Heads	8	16
Model Dimension	512	1024
Feed-Forward Dimension	2048	4096

GPT-1 kept only the decoder, and pretrained it

- Transformer Decoder with 12 layers
- BPE tokenizer of size 40k
- Trained on BookCorpus (~1B words)
- Pretrain, then fine-tune per task



GPT-1's recipe is the one you trained with this morning

	GPT-1, 2018	Your model
Batch	64 sequences of 512 tokens	64 windows of 512 tokens
Activation	GELU	GELU
Positions	learned	learned
Initialization	$N(0,0.02)$	$N(0,0.02)$
Schedule	warmup, then cosine decay	warmup, then cosine decay
Dropout	0.1	0.1
Norm placement	after each sublayer	before, like GPT-2
Size	768 wide, 12 layers, 12 heads	256 wide, 6 layers, 8 heads
Data	BookCorpus, 100 epochs	TinyStories, less than one pass

GPT-2 and GPT-3 mostly made it bigger

GPT-2 (2019)

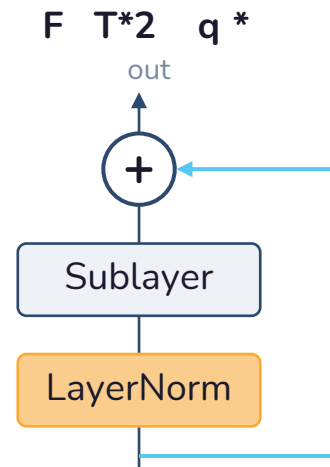
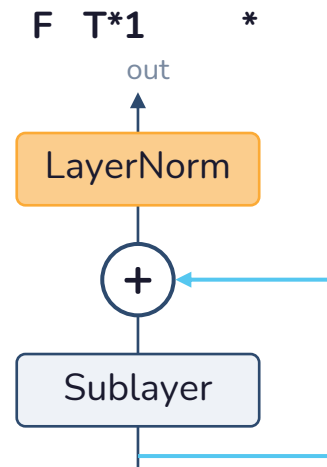
- Jump from 117M to 1.5B parameters and trained on a much larger dataset (40 GB)
- First model that showed good results in *zero-shot or few-shot prompting*
- Solve a task without fine-tuning, only prompt it with no or some examples
- And scaling seems to increase the performance

GPT-3 (2020), first closed model 🧐

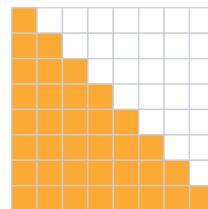
- Jump from 1.5B to 175B parameters and trained on 570 GB of data
- First model that performs exceptionally well with few-shot prompting
- *The end of the fine-tuning paradigm!?*

The two lines GPT-2 and GPT-3 did change

- **GPT-2:** layer normalization moved before each sublayer, plus one more after the last block. That is the pre-LN you implemented on Tuesday, and the final LayerNorm in your GPTModel1.
- **GPT-3:** the same model as GPT-2, except that its layers alternate between full attention and attention restricted to a local band.
- On Thursday, one pair rebuilds GPT-3's change as sliding-window attention.

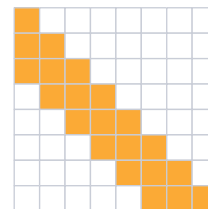


Your model is architecturally GPT-2. One pair on Thursday turns it into GPT-3.



full

alternating

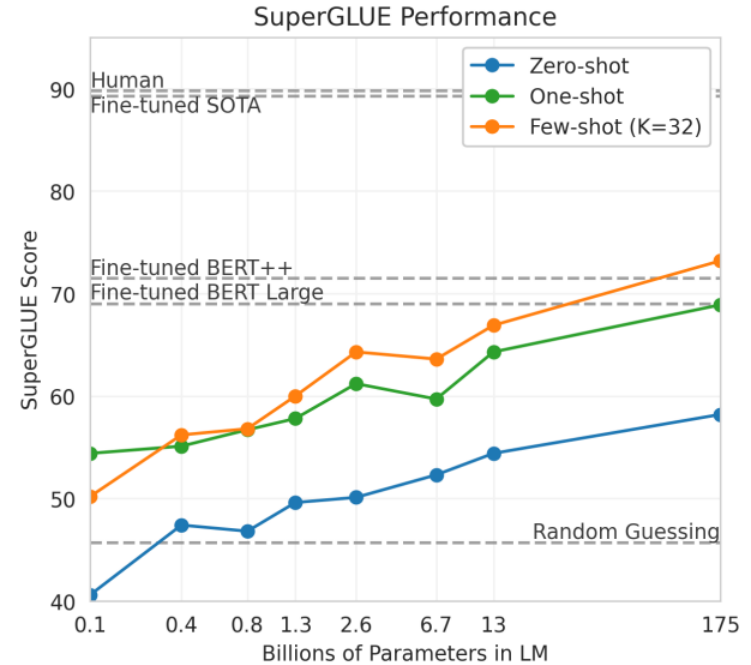
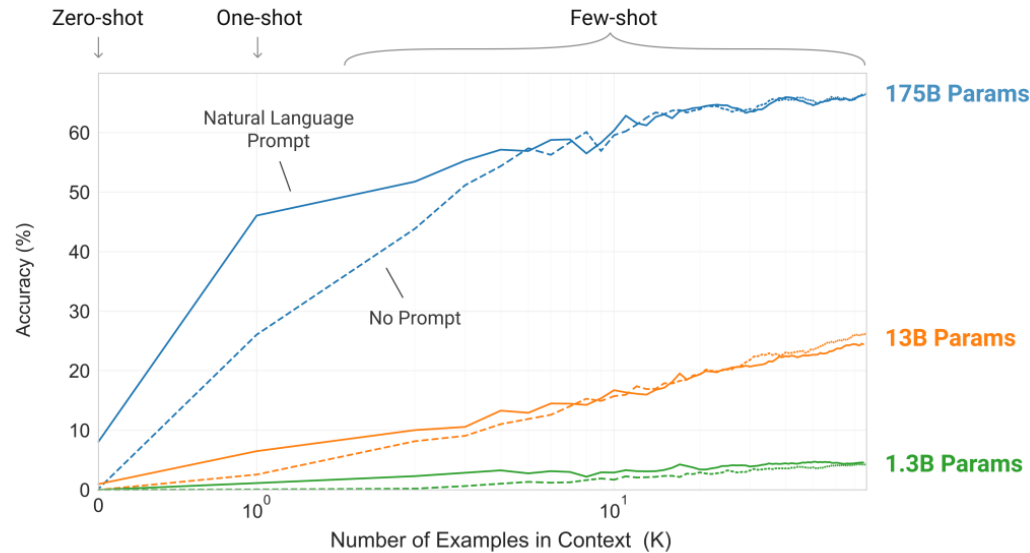


local band

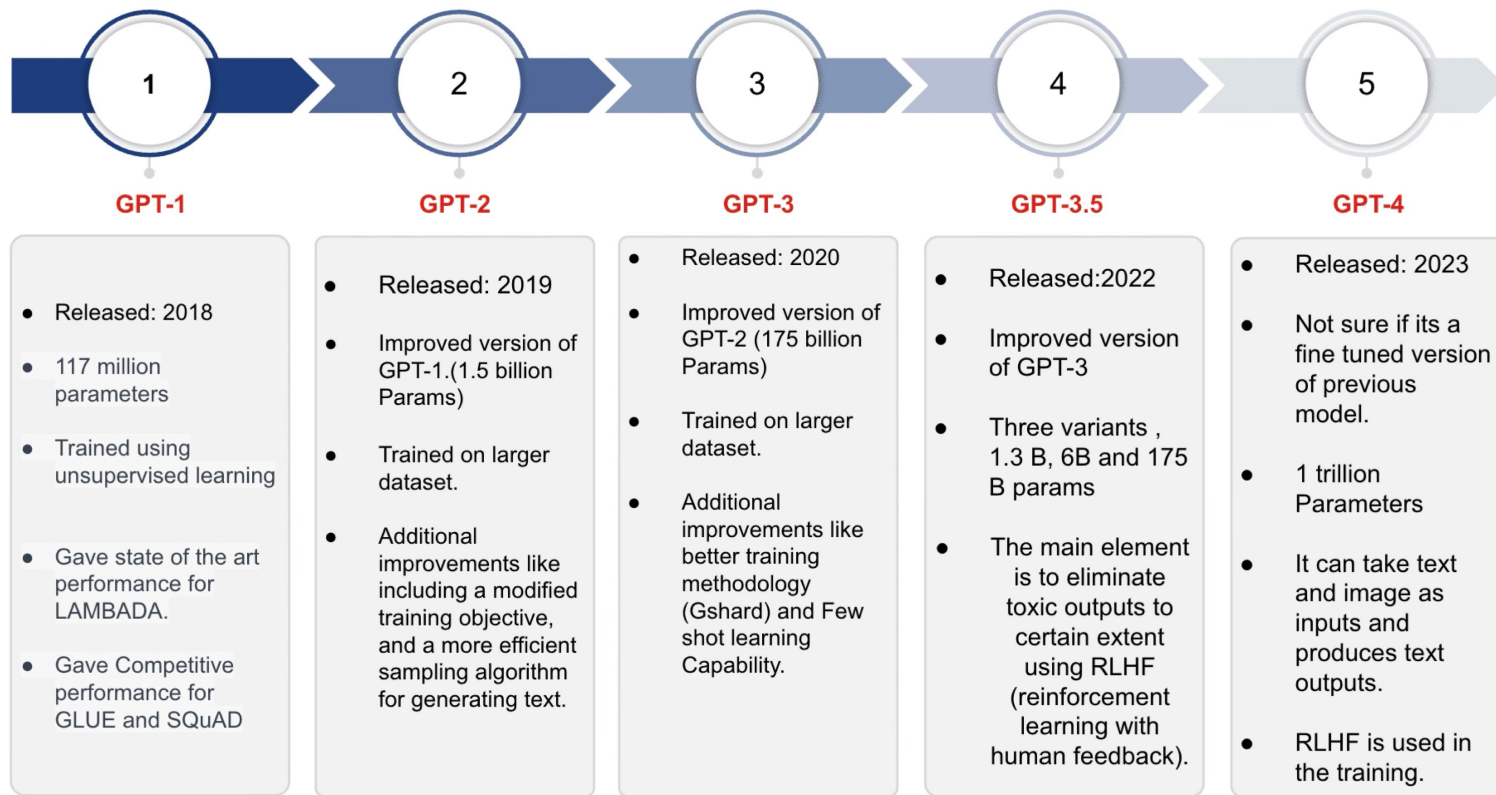
GPT-3 learns a task from examples in its prompt

- GPT-3 (175B) is far too big to fine-tune per task, since a gradient update on 175B parameters for every example is very expensive.
- **The shift:** put examples in the prompt.
- **Zero-shot:** task description only
- **One-shot / few-shot:** description plus one or a handful of examples
- This is just a forward pass: no gradient updates, no weight changes.
- Empirically few-shot beats one-shot beats zero-shot, and larger models use demonstrations better

The bigger the model, the more it gains from examples



GPT Development

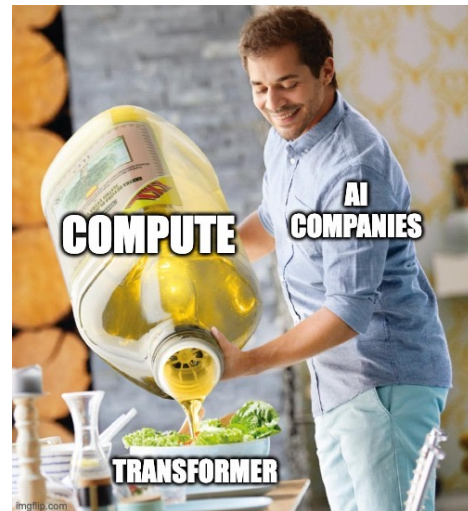


02.

Scaling, 2020 to 2022

After GPT-3, the rest was scaling

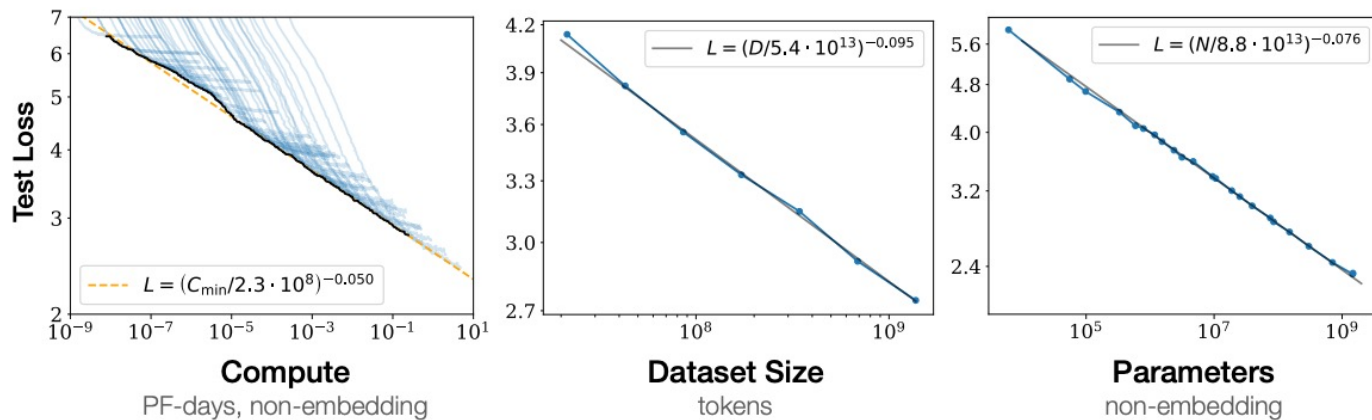
- As it turns out, the Transformer architecture is amazing for pre-training models on huge datasets
- And the Encoder (BERT etc) or Decoder (GPT etc) themselves make great Language Models
- The rest is scaling
(although attention has quadratic costs*)



*(two groups will work on that)

Loss falls predictably with size, data and compute

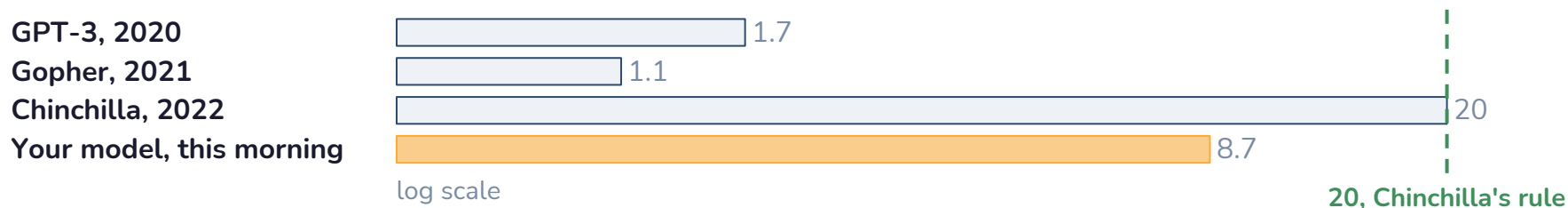
- Scaling payoff is predictable [Kaplan et al., 2020].
- Test loss falls as a smooth power law in model size, data, and compute
- As long as the other two are not the bottleneck, each improves the model along a straight line on a log-log plot. Bigger models are also more sample-efficient
- Chinchilla: GPT-3 was undertrained. For a fixed compute budget, prefer smaller models on more tokens, about 20 tokens per parameter.



GPT-3 was undertrained

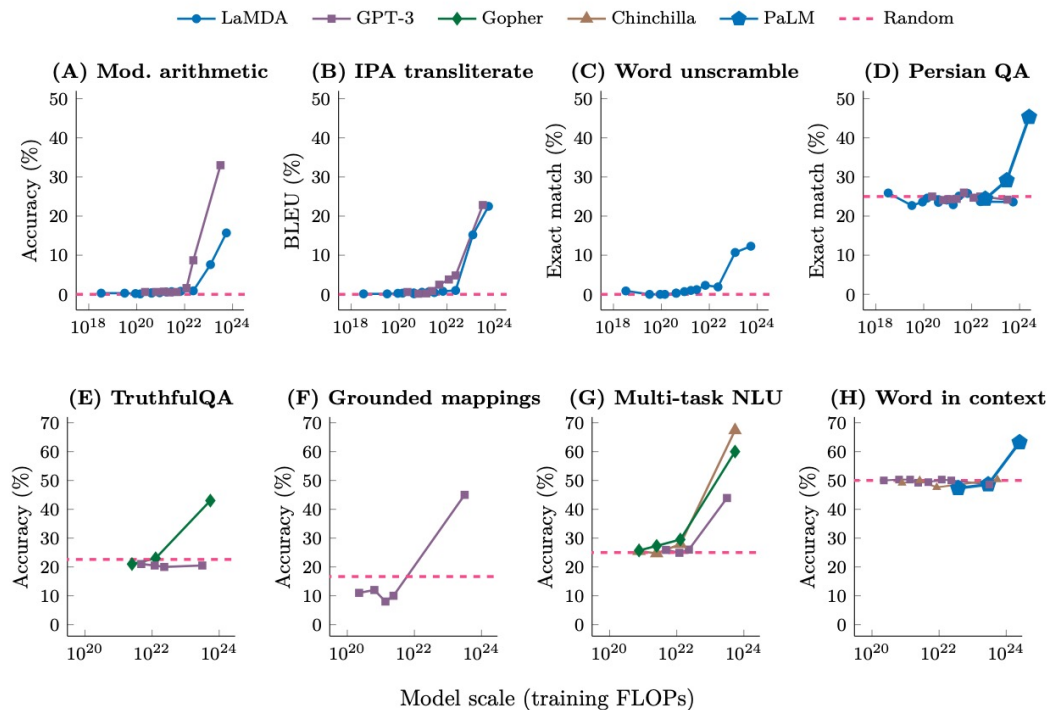
Model	Parameters	Tokens	Tokens per parameter
GPT-3, 2020	175B	300B	1.7
Gopher, 2021	280B	300B	1.1
Chinchilla, 2022	70B	1.4T	20
Your model, this morning	17.7M	153M	8.7

Chinchilla used the same compute as Gopher, a quarter of the parameters and four times the data, and beat Gopher and GPT-3.



Some abilities seem to appear suddenly at scale

- Some skills do not improve smoothly. They are near-random until a certain scale, then jump.
- Examples: multi-step arithmetic, word unscrambling, and chain-of-thought reasoning (working a problem out step by step)
- The same trick can fail at 1B parameters and work at 100B



Q **Iq** contested. Many of these jumps turn into smooth curves when measured with a continuous metric instead of all-or-nothing exact match (Schaeffer, Miranda, Koyejo 2023). Whether an ability “emerges” can depend on how you score it.

03.

Assistants, 2022

A 1.3B model trained on human feedback beat 175B GPT-3

- InstructGPT, 2022: GPT-3, fine-tuned in three steps.
- People preferred the answers of the 1.3B InstructGPT over those of 175B GPT-3, a model a hundred times larger.
- We cover this in detail this afternoon and you will build the fine-tuning step.

Scale made models knowledgeable.
Fine-tuning made them useful.

1 Fine-tune on answers written by people

2 Train a reward model on people's rankings of answers

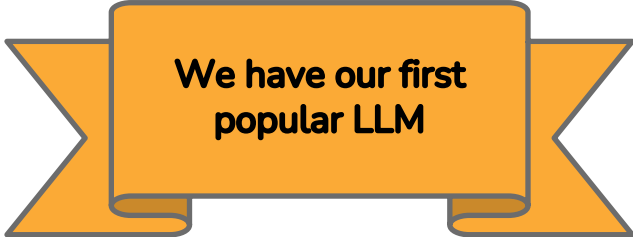
3 Optimize the model against that reward with reinforcement learning

ChatGPT is InstructGPT, made for dialogue

Same technique as InstructGPT with a focus on dialogue and using GPT 3.5

Methods

We trained this model using Reinforcement Learning from Human Feedback (RLHF), using the same methods as InstructGPT, but with slight differences in the data collection setup. We trained an initial model using supervised fine-tuning: human AI trainers provided conversations in which they played both sides—the user and an AI assistant. We gave the trainers access to model-written suggestions to help them compose their responses. We mixed this new dialogue dataset with the InstructGPT dataset, which we transformed into a dialogue format.

An orange ribbon graphic with a central rectangular box containing text. The ribbon has a 3D effect with a shadow.

**We have our first
popular LLM**

04.

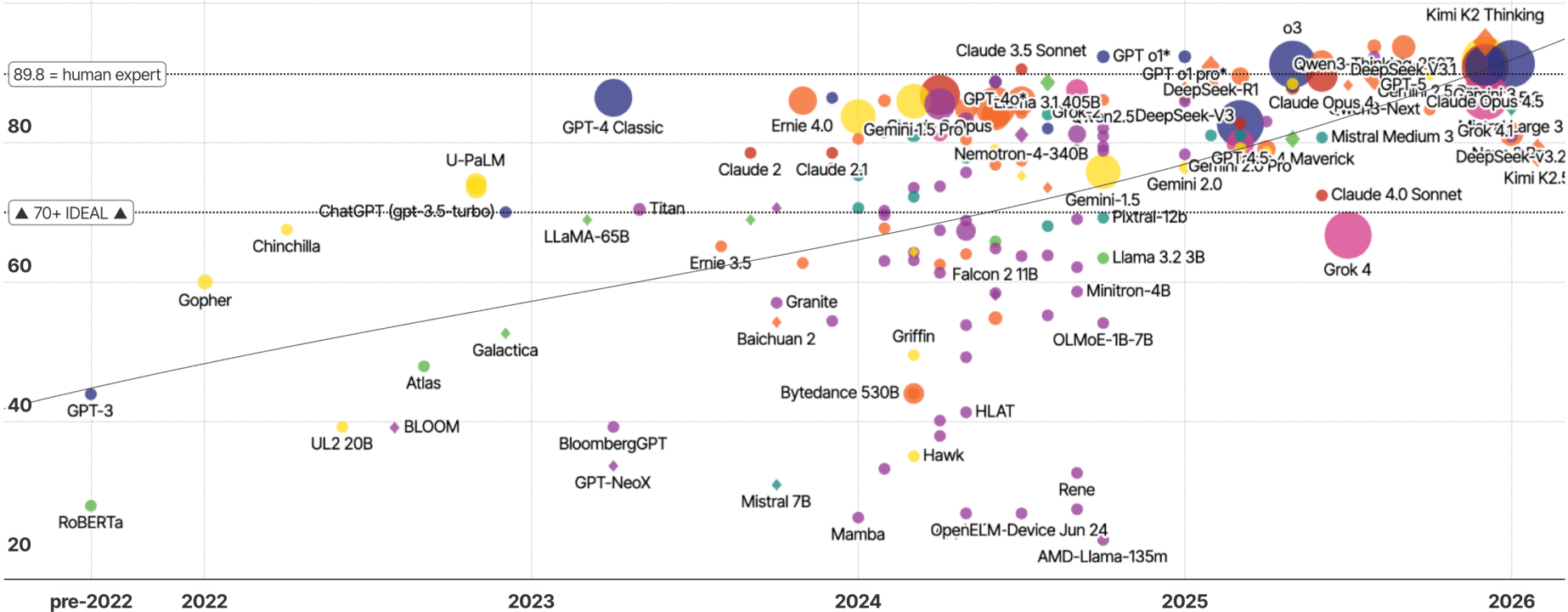
Today, 2023 to 2026

The LLM landscape, October 2026



🔍 search...

show only: all



Source: Information is beautiful

Three US labs, Chinese open weights, open models

- Three big US players: OpenAI (GPT), Google (Gemini), Anthropic (Claude)
- Chinese open-weights models
 - DeepSeek, Qwen, Kimi, Minimax, GLM
- Some ~~open-source~~ open-weights models
 - Llama family (Meta), GPT OSS (OpenAI), Mistral, Gemma (Google), Phi (Microsoft)
- Open Source: training data, algorithms and weights are available
 - OLMo (AI2, Seattle), T5 (Google), Apertus (Swiss AI, ETH Zürich), SmolLM 🤗
 - Nemotron (Nvidia), Soofi S (KI Bundesverband etc, based on Nemotron)
 - Kolibri (Aleph Alpha)

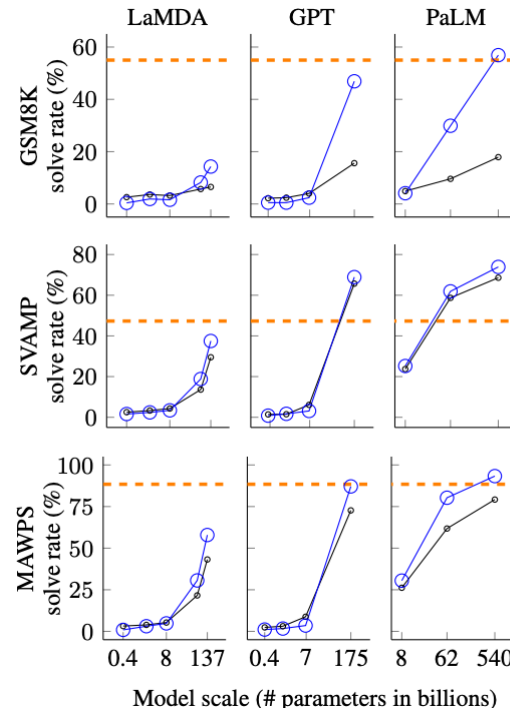
Since GPT-2, the changes are Thursday's components

GPT-2, and your model	2026 open models	Thursday
LayerNorm	RMSNorm	01
Pre-LN	pre-LN, with post-LN variants back in OLMo 2	01
GELU feed-forward	SwiGLU, often as a mixture of experts	02
Full causal attention	sliding window in some layers	03
Raw query-key scores	QK-Norm	04
Learned positions	RoPE	05, 06
Multi-head attention	grouped-query attention	07
Generation without a cache	KV cache	08
Dropout 0.1	no dropout	

Asking for the steps first helps on hard problems

- Idea: make the model show its work before answering.
On multi-step problems this helps a lot.
- Zero-shot CoT: just add "Let's think step by step."
- Few-shot CoT: include worked examples that show the reasoning, not just the answer
- Self-consistency: sample several reasoning paths and take the majority answer. Aggregating beats any single path

Standard Prompting	Chain-of-Thought Prompting
Model Input Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now? A: The answer is 11. Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?	Model Input Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now? A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11. Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?
Model Output A: The answer is 27. ❌	Model Output A: The cafeteria had 23 apples originally. They used 20 to make lunch. So they had $23 - 20 = 3$. They bought 6 more apples, so they have $3 + 6 = 9$. The answer is 9. ✅



Reasoning models learned from checkable rewards

- DeepSeek-R1, January 2025: start from a base model and train it with reinforcement learning, no fine-tuning on examples first.
- The reward is a rule, not a learned model: is the final answer correct and is it in the right format.
- On a hard math benchmark, accuracy rose from 15.6 to 71.0 percent, and the answers grew longer on their own: the model learned to spend more tokens thinking.
- The authors avoided learned reward models on purpose: at this scale, the model learns to game them.

A qJ IJ IJ 9 40 IJ IJq IJ 11 15
v f q

Answer: 1 h 35 min

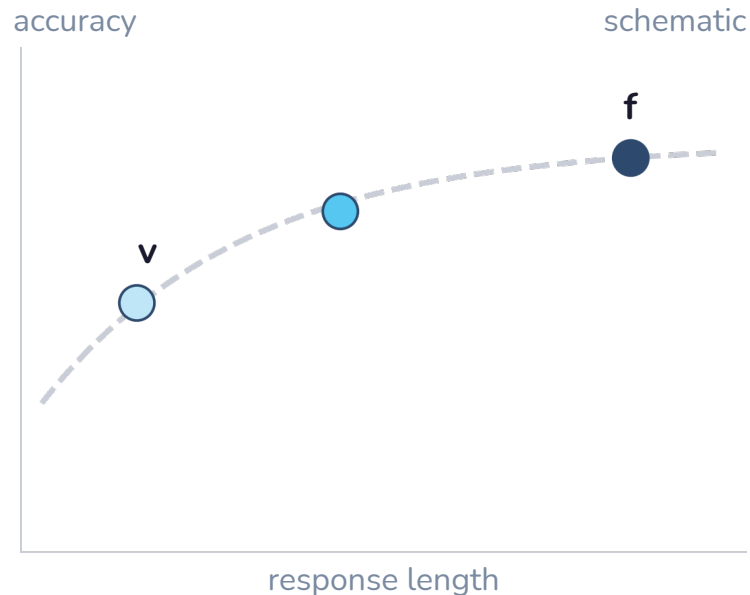
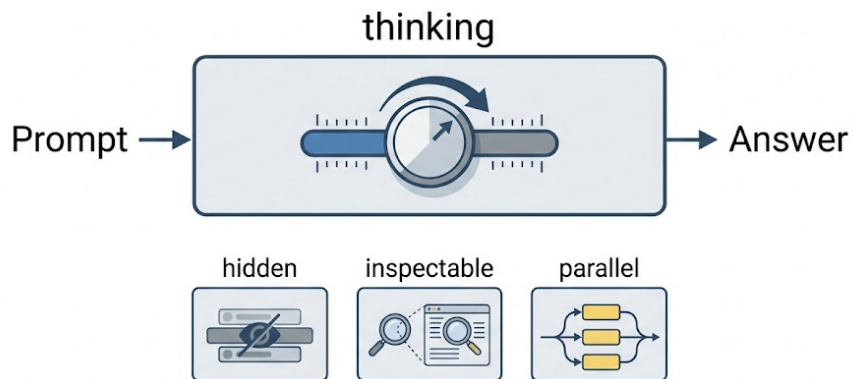
short answer

From 9:40 to 10:00 is 20 minutes.
From 10:00 to 11:00 is 60 minutes.
From 11:00 to 11:15 is 15 minutes.
 $20 + 60 + 15 = 95$ minutes.
Check: 9:40 plus 95 minutes is 11:15.
Correct.
Answer: 1 h 35 min

reasoning trace, then the same answer

Thinking costs tokens, and you choose how many

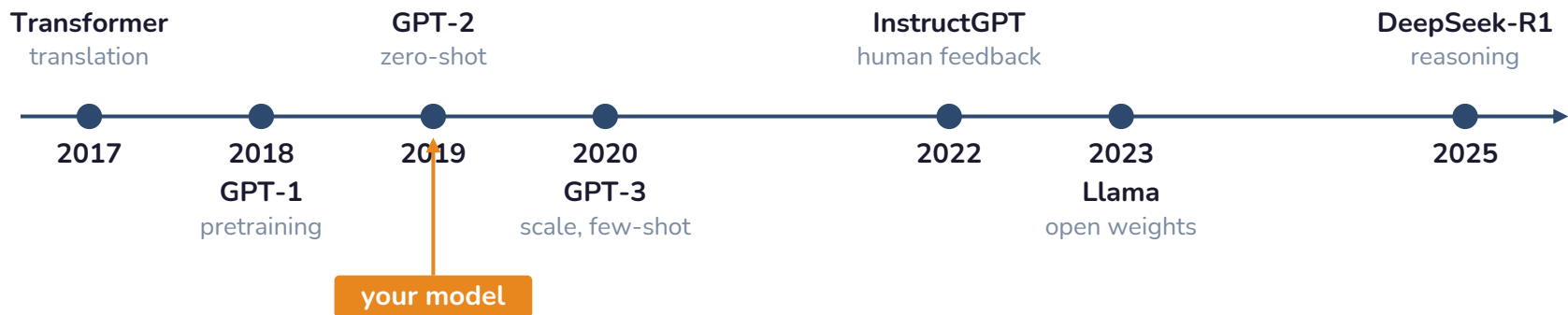
- Reasoning models write out their reasoning before the answer. Longer reasoning, better answers, more tokens and more time.
- Most models take a reasoning effort of low, medium or high in their instructions.
- The trade-off is now a setting: spend compute at training time, or at answer time.
- Where the thinking lives varies: hidden tokens, an inspectable budgeted block, or parallel hypotheses



Every LLM since GPT-1 is your model, with a few lines changed.

This afternoon, you do what InstructGPT did

- 2017 to 2025 on one line: translation, pretraining, scale, human feedback, open weights, reasoning.
- Your model has reached 2019: GPT-2's architecture, pretrained.
- This afternoon it takes the 2022 step: fine-tuning on instructions, then preferences.



Sources

Literature

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Literature, continued

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